central office called the Mobile Telephone Switching Office (MTSO). This office handles all of the phone connections to the normal land-based phone system, and controls all of the base stations in the region.

Now that we have a fair idea about the setup of a cellular network, let's see how a call is placed.

When you turn on the phone, the phone tries to communicate with a specific channel called the control channel. The phone receives a unique identity number called System Identification Code (SID, specific to each network operator) and compares it to the SID programmed into the phone. If they match, it knows that it's using the home network, if it doesn't, it's on roaming. If the phone can't find any control channels to listen to, it knows it's out of range and displays a "no service" message.

It also sends a registration request. This way, the MTSO can keep a track of your location in its database. It looks for you in the database when you get a call.

The MTSO then selects a frequency pair on which you'll communicate and using the control channel tells your phone to use these frequencies. Once your phone and the tower switch on those frequencies, the call is connected.

If you're travelling you're bound to change cells. As you move closer to the edge of a cell, your signal strength decreases, meanwhile the tower in the other cell (the one you're moving closer to) notices that your phone signal strength is increasing. These two towers coordinate with the MTSO and it records that you've switched cells.

There are three technologies that are used by network operators to carry out this communication. Although the names look scary, you can break it down to simple parts. The first word tells you what the access method is. The second word, division, lets you know that it splits calls based on that access method. The last part says multiple access which means that more than one user can utilise each cell

- **Frequency division multiple access (FDMA):** FMDA puts each call on a different frequency. It separates the spectrum into distinct voice channels by splitting it into uniform chunks of bandwidth. It's used for analog transmission but is not considered an efficient method for digital transmission.
- **Time division multiple access (TDMA):** In TMDA, a narrow band (channel) that is 30 kHz wide and 6.7 milliseconds long is split time-wise into three time slots. Each conversation gets the radio for one-third of the time. This is possible because voice data that has been converted to